Trend Analysis of COVID-19 Waves and their Geographic Spread

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STAT 674 – Applied Database Management

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Background - Rahul

Talk about COVID-19 background, first case when?, impact roughly on the world

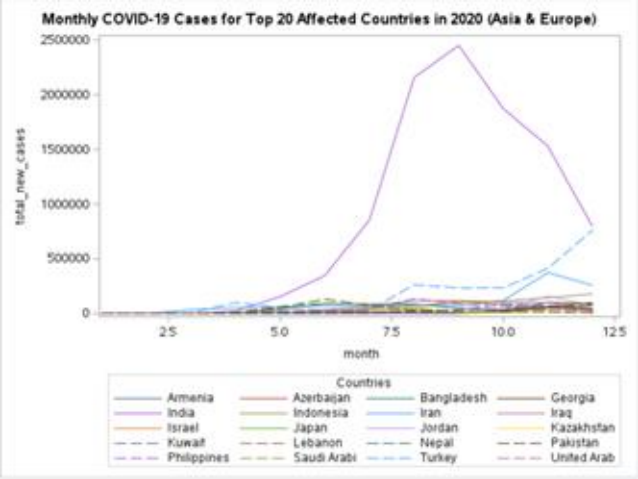
Goal of Project - Rahul

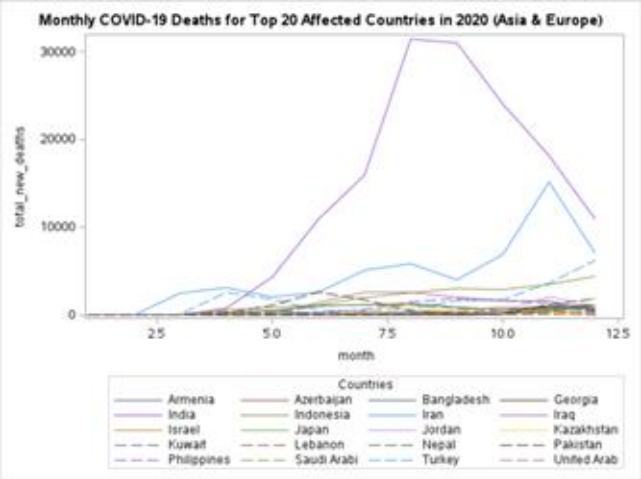
General goal of project – wrote on project milestone

Goal of project pt. 2

Digging into the goal – maybe we have the table on which continents first reported cases

Maybe the top 20 summary graphs could go here too

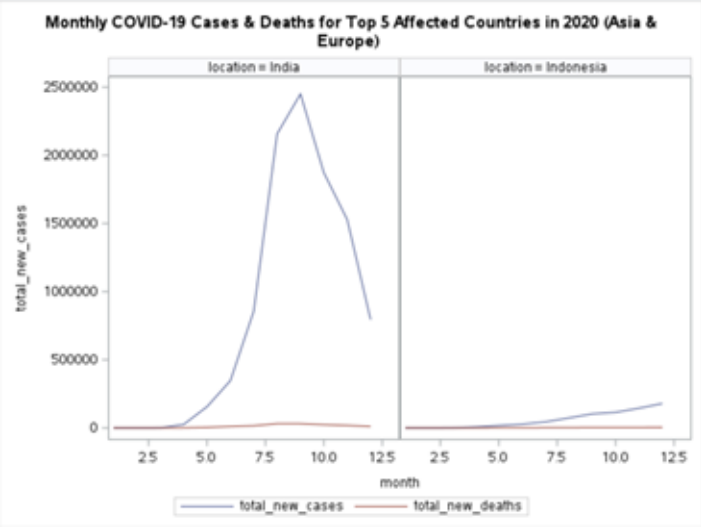


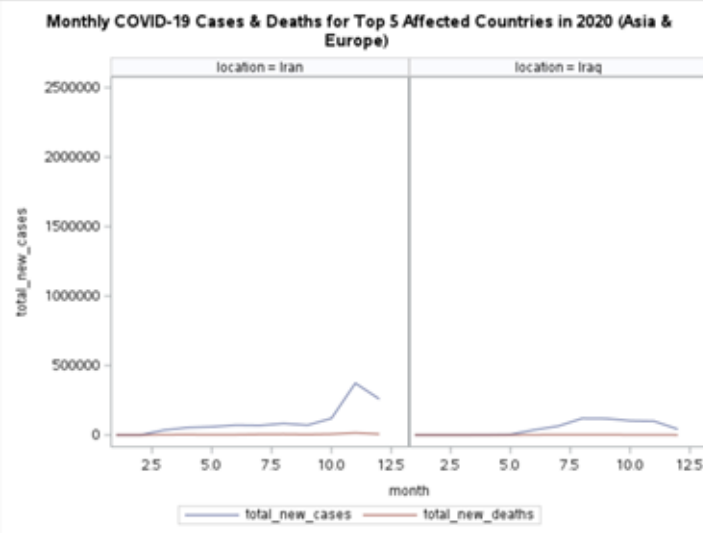


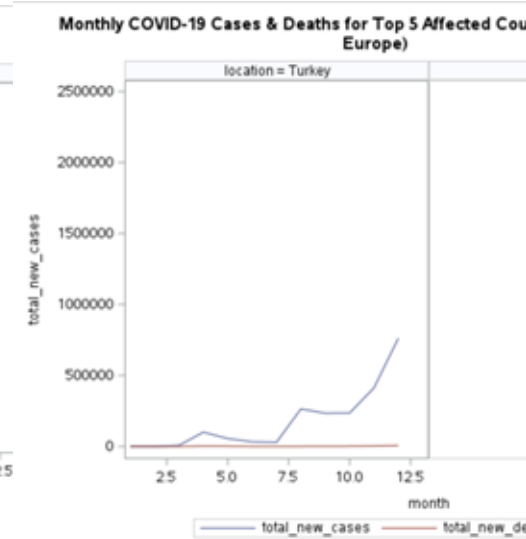
Core SAS Procedurals Used - Rahul

- Proc Sql
- Proc SgPanel
- Proc PgPlot

Top 5 Countries in Asia and Europe with the Highest Number of Reported Cases

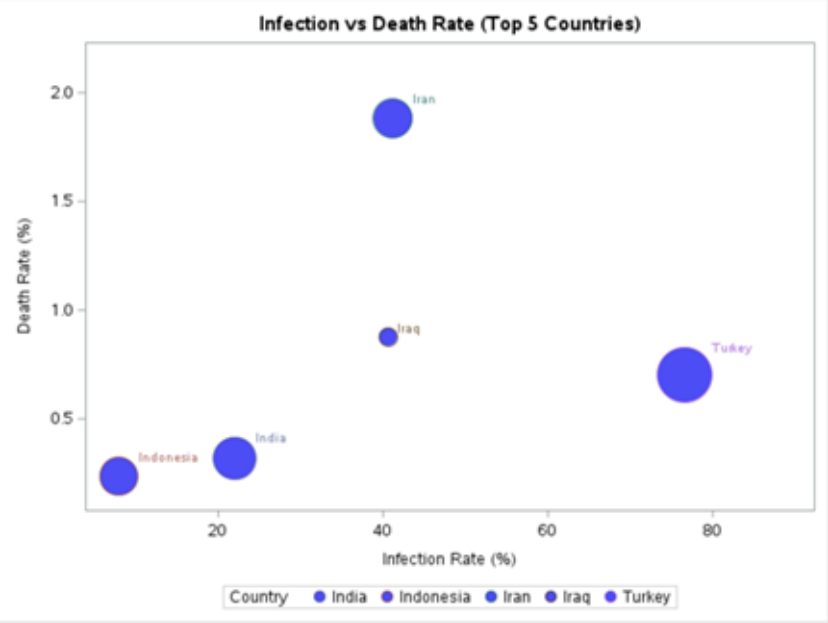


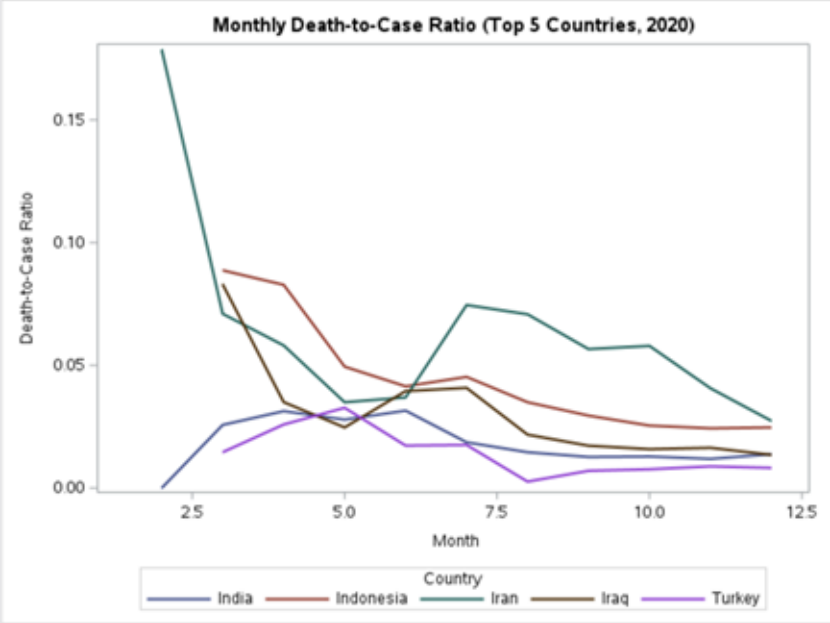




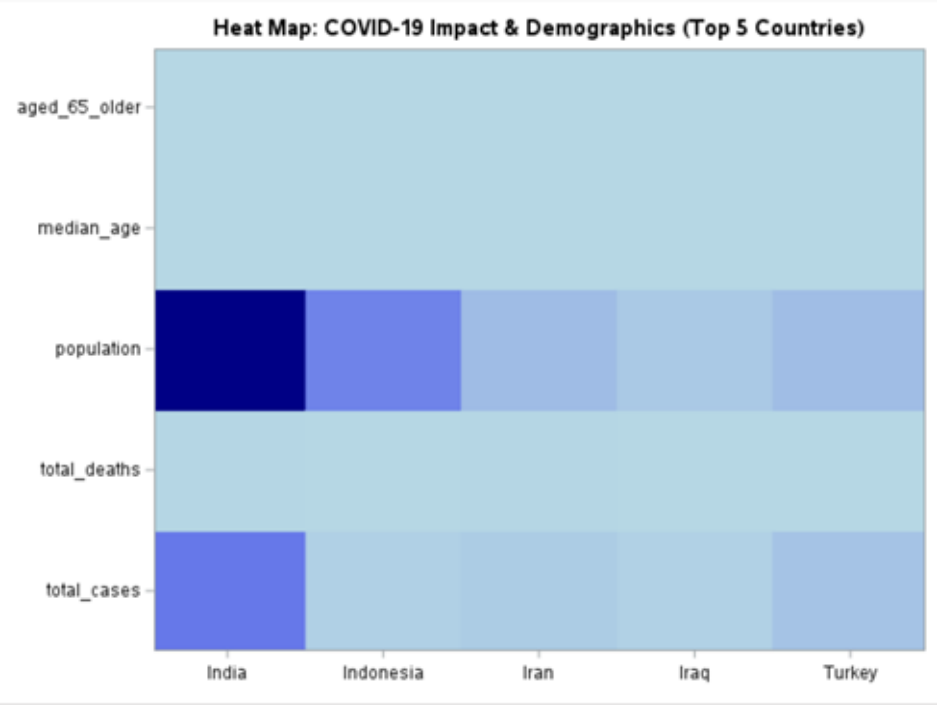
Bubble Plot, Monthly Death to Case Ratio

Interpret these graphs



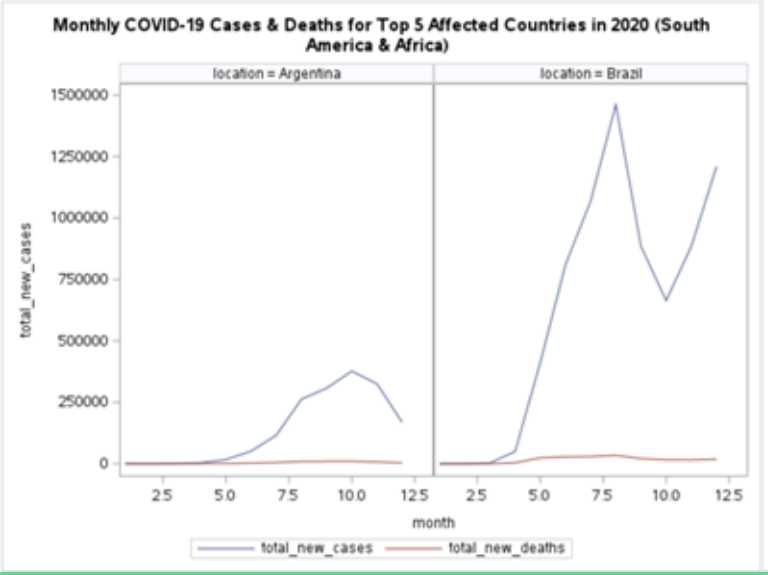


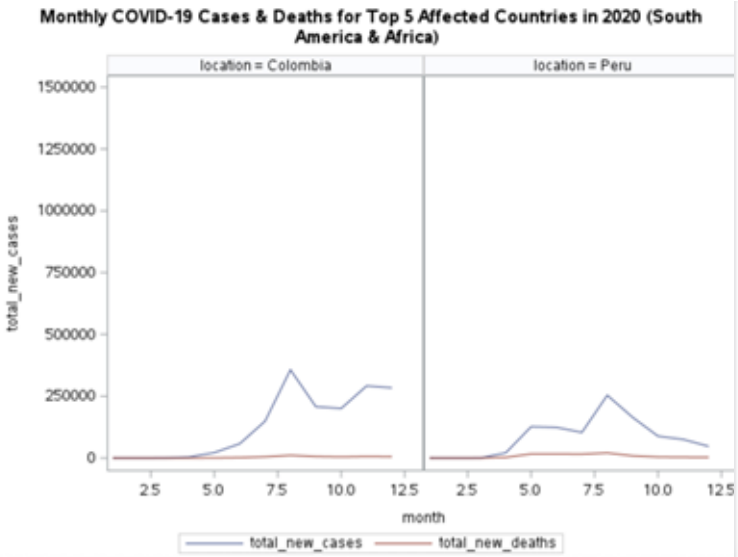
Heat Map?

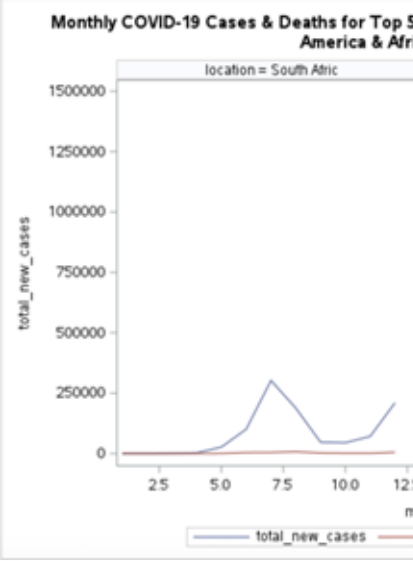

Heat Map: COVID-19 Impact & Demographics (Top 5 Countries)
aged_65_older
median_age
population
total_deaths
total_cases
India
Indonesia
Iran
Iraq
Turkey

Top 5 Countries in South America & Africa

- Brazil, Argentina, Colombia, Peru, and South Africa were the top 5 countries within South America and Africa with the highest number of reported cases throughout 2020
- Brazil had the highest number of cases at 7,448,560 and the highest number of deaths at 190,488
- No obvious correlation between reported cases and number of deaths

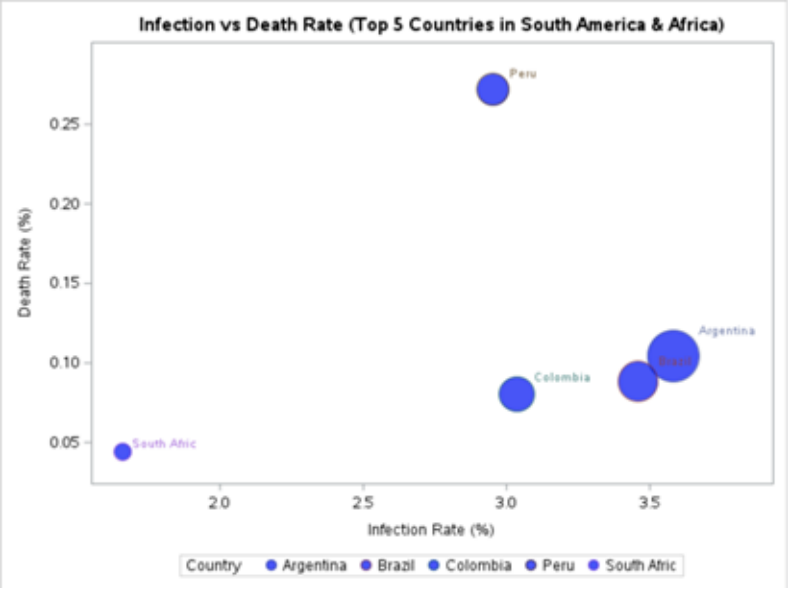


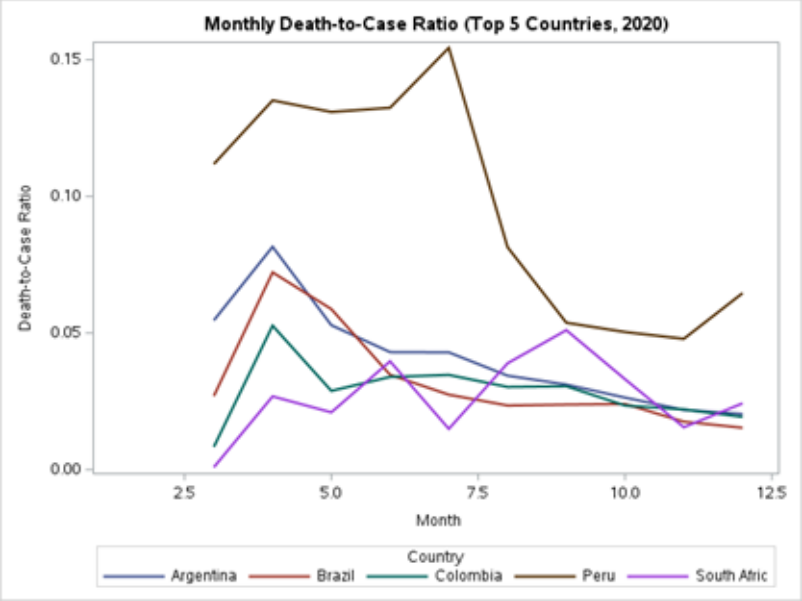




Further Analysis on South America & Africa

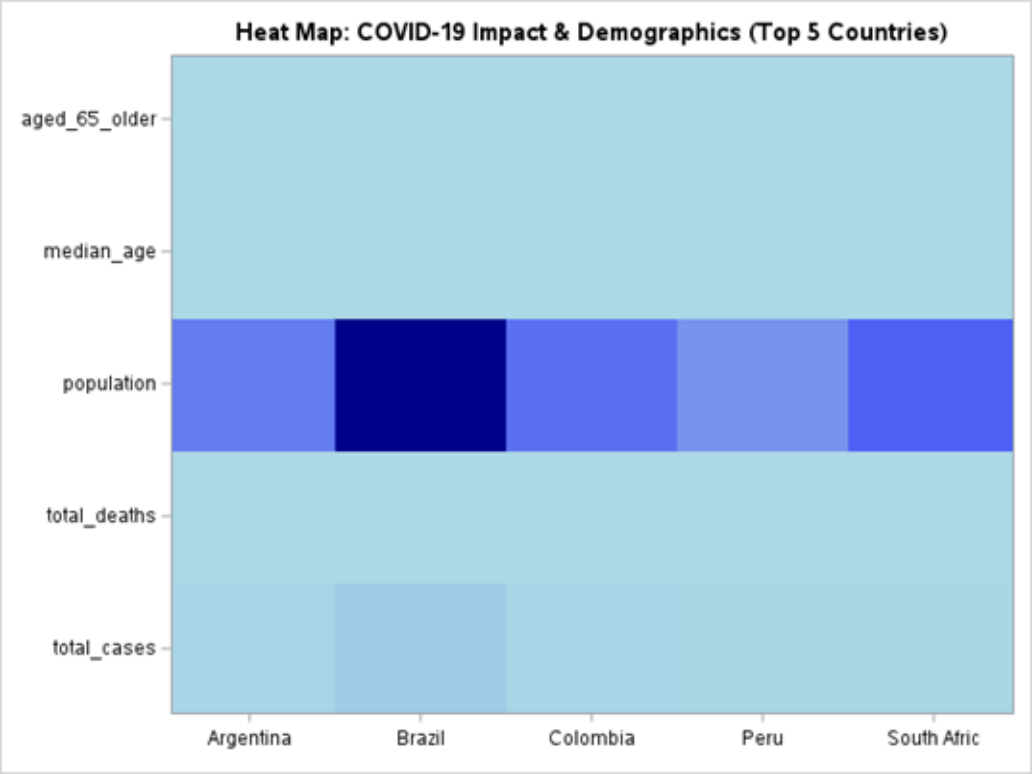
- While many countries had the most deaths in comparison to COVID cases in the beginning of 2020, Peru's death-to-case ratio peaked mid-year
- Peru had a significantly higher death-to-case ratio than the other four countries





Heat Map Analysis

- Brazil has a significantly higher population, which corresponds to their higher number of total cases
- No significant differences between 65 and older, median age, or total deaths amongst



Conclusions and Findings